Life Insurance Analysis:
Group Assignment for Life Insurance Course
Personalization 3-1-1-2

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1 Introduction

In this report, I develop and analyze a pension scheme specifically designed for Mr. and Ms. Bünzli, a Swiss couple approaching retirement. Unlike traditional pension models, this scheme integrates additional features aimed at meeting the unique financial needs and goals of retirees. The analysis focuses on determining the actuarial present value (APV) of the pension scheme and examining how the incorporation of these features can enhance the overall financial security for Mr. and Ms. Bünzli. Through a detailed examination of various financial and actuarial assumptions, I aim to provide a comprehensive evaluation of the pension scheme's effectiveness. The ultimate goal is to offer insights into whether such a scheme could serve as a viable retirement planning solution, considering both its potential benefits and limitations. In the first part I will state the assumptions, followed by the calculations for the basic product and last the calculations for the personalized product.

2 Assumptions

In order to proceed with a structured and accurate analysis, I have established a set of key assumptions regarding the couple, the characteristics of the pension scheme, and the underlying financial environment:

2.1 About the Pension Scheme and Beneficiaries

1. Mr. and Ms. Bünzli, the beneficiaries, are both 60 years old, planning to retire in 5 years.

2. Their life expectancies are based on the national life table provided by the Bundesamt für Statistik in Switzerland (2024), reflecting typical mortality rates. Female table for Ms. Bünzli, male table for Mr. Bünzli. The sources for these life tables are retrieved from the Bundesamt für Statistik website.

3. The pension scheme is configured as a joint life annuity, which will cease with the first death among the couple.

4. Payments from the pension are structured to occur at three specific times each year: at the beginning, seven months in, and finally, eleven months after the year commences, with the first payment set at 12,000 CHF.

5. To account for inflation and increase the pension's value over time, the payments will increase by 1% annually.

6. The couple aims to accumulate 200,000 for retirement through a pure endowment policy with monthly premiums payable until one month before retirement, this pure endowment is bought by Ms. Bünzli.

2.2 Financial Assumptions

7. For the purpose of discounting, we will apply a fixed annual interest rate of 2.320%. This rate is derived from the 5-year fixed mortgage offering by Zürcher Kantonalbank (2024), recognized for being the highest-rated bank in Switzerland in terms of creditworthiness. The choice of a 5-year period is strategic, aligning with the timeline until the couple's retirement. This rate reflects the prevailing market conditions and is considered an appropriate proxy for the risk-free interest rate, especially in light of the Swiss housing market's notable stability.

8. A uniform distribution of deaths assumption is applied for fractional age calculations, simplifying the mortality rate projections used in this study.

Utilizing these assumptions, the subsequent sections will delve into the calculation of the endowment policy's monthly premium and the APV of the pension scheme. The findings and insights derived from this analysis aim to provide a solid foundation for making informed decisions regarding retirement planning for Mr. and Ms. Bünzli.

3 Calculations

3.1 Basic product

A key component of the couple's retirement strategy involves accumulating CHF 200,000 through a pure endowment policy, featuring monthly premiums that are payable up until one month before retirement. Let A denote said monthly premium. The product is bought by Ms. Bünzli.

3.1.1 monthly premium payment

$$Y = A * \ddot{a}^{(12)}_{60:\overline{5}|}$$

$$= A \times \sum_{k=0}^{\min\{K^{(12)}_{60},59\}} \left(\frac{1}{v}\right)^{\frac{60-k}{12}} = A \times \left(\frac{1}{v}\right)^{5} \times \sum_{k=0}^{\min\{K^{(12)}_{60},59\}} v^{\frac{k}{12}}$$

$$E[Y] = A\left(\frac{1}{v}\right)^{5} \times \frac{1 - E[v^{\min\{K^{(12)}_{60}+\frac{1}{12},5\}}]}{1 - v^{\frac{1}{12}}} = A\left(\frac{1}{v}\right)^{5} \times \frac{1 - A^{(12)}_{60:\overline{5}|}}{1 - v^{\frac{1}{12}}}$$

This value mus be equal to 200'000.

$$A\left(\frac{1}{v}\right)^{5} \times \frac{1 - A^{(12)}_{60:\overline{5}|}}{1 - v^{\frac{1}{12}}} \overset{!}{=} 200000 \Leftrightarrow \frac{200000(1 - v^{\frac{1}{12}})v^{5}}{1 - A^{(12)}_{60:\overline{5}|}} = A$$

$$A^{(12)}_{60:\overline{5}|} = \sum_{k=0}^{59} v^{\frac{k+1}{12}}\, {}_{\frac{k}{12}|\frac{1}{12}}q_{60} + v^{5}\,{}_{5}p_{60} \overset{\text{UDD}}{=} 0.8925939 \Rightarrow A = 3170.299$$

3.1.2 Variance

$$VAR[Y] = \text{VAR}\left[A\left(\frac{1}{v}\right)^5 \frac{1 - v^{\left\{\min\left\{K_{60}^{(12)} + \frac{1}{12}, 5\right\}\right\}}}{1 - v^{\frac{1}{12}}}\right]$$

$$= \left(\frac{A\left(\frac{1}{v}\right)^5}{1 - v^{\frac{1}{12}}}\right)^2 VAR\left[v^{\left\{\min\left\{K_{60}^{(12)} + \frac{1}{12}, 5\right\}\right\}}\right]$$

$$= \left(\frac{A\left(\frac{1}{v}\right)^5}{1 - v^{\frac{1}{12}}}\right)^2 \left({}^2A_{60:\overline{5}|}^{(12)} - (A_{60:\overline{5}|}^{(12)})^2\right)$$

$${}^2A_{60:\overline{5}|}^{(12)} = \sum_{k=0}^{59} v^{\frac{2(k+1)}{12}} {}_{\frac{k}{12}|\frac{1}{12}}q_{60} + v^{2*5} {}_5p_{60} \overset{\text{UDD}}{=} 0.7968197 \Rightarrow VAR = 332644927$$

3.1.3 Comment

The monthly payments were found to be 3170.299 this is a rather high value especially if one compares it to the Swiss median income of 6665 per month (Bundesamt für Statistik, 2020). Furthermore, one has to acknowledge the fact that it is a pure endowment and the probability of Ms. Bünzli not reaching the age 65 is 2.02% in which case the money would be gone and the spouse would be left with nothing. If I were to advise Mr. and Ms. Bünzli I would not suggest this product.

3.2 personalized Product

This is a geometrically increasing joint life annuity for a retiring couple at 65, starting with 12,000 in the first year (4,000 per payment at the start, 7th, and 11th months), increasing 1% annually until the first partner's death. Let ${}_{k|t}q_x$ be the probability that both survive until k and then one dies within the following t years. Furthermore, it is assumed that both will reach retirement age. T_x is the time of death of the first person of the couple.

3.2.1 APV

$$\begin{aligned} APV = 4000E[\ & \sum_{k=0}^{\infty} v^k (1.01)^k \mathbf{1}_{\{T_x \geq k\}} \\ & + \sum_{k=0}^{\infty} v^{(k+\frac{11}{12})} (1.01)^k \mathbf{1}_{\{T_x \geq k+\frac{7}{12}\}} \\ & + \sum_{k=0}^{\infty} v^{(k+\frac{7}{12})} (1.01)^k \mathbf{1}_{\{T_x \geq k+\frac{11}{12}\}}\ ] \end{aligned}$$

Let frac denote the fractional part of the year, specifically 0, $\frac{7}{12}$, or $\frac{11}{12}$, which accounts for payments occurring at different times within the year. Let s denote the corresponding survival period, which varies based on the age and the fraction of the year under consideration.

$$APV = 4000 \times \sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} v^{j+\text{frac}} \times (1.01)^j \right) \times {}_{k+\text{frac}|s}q_{65} \overset{\text{UDD}}{=} 67619.32$$

3.2.2 Variance

$$VAR = E[Y^2] - E[Y]^2$$

$$E[Y^2] = 4000^2 \times \left(\sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} v^{j+\text{frac}} \times (1.01)^j \right)^2 \times {}_{k+\text{frac}|s}q_{65} \right)$$

$$\Rightarrow VAR = 4000^2 \left[E[Y^2] - E[Y]^2 \right] \overset{\text{UDD}}{=} 6679926596$$

4 References

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